Based Material Line Up

S1000-2/S1000-2B

1. CORE (C-STAGE)

Thickness		ply-up	RC (%)	Dk				Df			
mm	mil			1 GHz	3 GHz	5 GHz	10 GHz	1 GHz	3 GHz	5 GHz	10 GHz
0.050	2.00	1x1067	64	4.12	4.04	3.99	3.95	0.019	0.019	0.020	0.021
0.063	2.52	1x1067	70	3.97	3.87	3.82	3.78	0.021	0.021	0.023	0.022
0.076	3.04	1x1080	64	4.12	4.04	3.99	3.95	0.019	0.019	0.020	0.021
0.090	3.60	1X3313	50	4.46	4.44	4.39	4.34	0.017	0.017	0.019	0.017
0.100	4.00	1x3313	56	4.31	4.27	4.22	4.17	0.018	0.018	0.020	0.018
0.100	4.00	2x106	70	3.97	3.87	3.82	3.78	0.021	0.021	0.023	0.022
0.110	4.40	1x2116	49	4.49	4.45	4.42	4.38	0.016	0.016	0.017	0.018
0.110	4.40	2x106	74	3.87	3.76	3.70	3.66	0.022	0.022	0.024	0.023
0.125	5.00	1x2116	55	4.34	4.29	4.25	4.21	0.017	0.018	0.018	0.019
0.150	6.00	1x1506	44	4.61	4.59	4.56	4.52	0.015	0.015	0.016	0.017
0.156	6.24	1x1506	45	4.59	4.57	4.53	4.49	0.015	0.015	0.016	0.017
0.200	8.00	1x7628	45	4.59	4.57	4.53	4.49	0.015	0.015	0.016	0.017
0.200	8.00	2x3313	56	4.31	4.27	4.22	4.17	0.018	0.018	0.020	0.018
0.240	9.60	2x2116	54	4.36	4.32	4.27	4.23	0.017	0.018	0.019	0.018
0.300	12.00	2x1506	44	4.61	4.60	4.56	4.51	0.015	0.016	0.017	0.015
0.350	14.00	2x7628	41	4.70	4.69	4.65	4.58	0.015	0.018	0.019	0.011
0.380	15.00	2x7628	43	4.64	4.63	4.59	4.54	0.015	0.016	0.017	0.015
0.400	16.00	2x7628	45	4.59	4.57	4.53	4.49	0.015	0.015	0.016	0.017
0.450	18.00	2X7628+1080	46	4.57	4.55	4.5	4.46	0.015	0.015	0.016	0.017

0.500	20.00	2X7628+2116	46	4.57	4.55	4.5	4.46	0.015	0.015	0.016	0.017
0.600	24.00	3X7628	43	4.64	4.63	4.59	4.54	0.015	0.016	0.017	0.015
0.710	28.00	4X7628	42	4.67	4.67	4.62	4.56	0.015	0.017	0.018	0.013
0.800	32.00	4X7628	43	4.64	4.63	4.59	4.54	0.015	0.016	0.017	0.015
1.000	40.00	5X7628	43	4.64	4.63	4.59	4.54	0.015	0.016	0.017	0.015
1.200	47.00	6X7628	43	4.64	4.63	4.59	4.54	0.015	0.016	0.017	0.015

2. PREPREG (B-STAGE)

Glass style	RC (%)	Thickness		Dk				Df			
	Nominal	mm	mil	1GHz	3GHz	5GHz	10GHz	1 GHz	3 GHz	5 GHz	10 GHz
1067	74	0.072	2.83	3.87	3.76	3.70	3.66	0.022	0.022	0.024	0.023
106/1037	70	0.050	1.97	3.97	3.87	3.82	3.78	0.021	0.021	0.023	0.022
	73	0.055	2.17	3.90	3.79	3.73	3.69	0.021	0.022	0.023	0.022
	76	0.063	2.48	3.82	3.70	3.65	3.61	0.022	0.022	0.024	0.023
	78	0.070	2.76	3.77	3.65	3.59	3.55	0.022	0.023	0.024	0.024
1080/1078	62	0.076	2.99	4.17	4.10	4.05	4.00	0.019	0.019	0.021	0.020
	64	0.078	3.07	4.12	4.04	3.99	3.95	0.019	0.019	0.020	0.021
	65	0.080	3.15	4.09	4.01	3.96	3.92	0.020	0.020	0.022	0.020
	68	0.089	3.50	4.02	3.93	3.88	3.83	0.020	0.021	0.022	0.021
2313	54	0.100	3.94	4.36	4.32	4.27	4.23	0.017	0.018	0.019	0.018
	57	0.106	4.17	4.29	4.24	4.19	4.14	0.018	0.018	0.020	0.018
2116	53	0.122	4.80	4.39	4.35	4.30	4.26	0.017	0.018	0.019	0.017
	56	0.130	5.12	4.31	4.27	4.22	4.17	0.018	0.018	0.020	0.018

	58	0.140	5.51	4.27	4.21	4.16	4.11	0.018	0.019	0.020	0.019
1506	44	0.150	5.91	4.61	4.60	4.56	4.51	0.015	0.016	0.017	0.015
	49	0.175	6.89	4.49	4.46	4.42	4.37	0.016	0.017	0.018	0.016
7628	43	0.193	7.60	4.64	4.63	4.59	4.54	0.015	0.016	0.017	0.015
	45	0.200	7.87	4.59	4.58	4.53	4.48	0.015	0.016	0.017	0.015
	48	0.216	8.50	4.51	4.49	4.44	4.40	0.016	0.017	0.018	0.016
	50	0.222	8.74	4.46	4.44	4.39	4.34	0.017	0.017	0.019	0.017

3. REMARK

1) Test by SPDR method.
2) The data above show actual values and are not guaranteed, for your reference only.
3) Last update: September, 2020.